

## CS486 MOCK FINAL EXAMINATION - TEST 2

Based on the FINAL REVIEW structure; total: 10 points

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### Question 1 - ER Models

[1 pt]

A hospital records **Departments**, **Doctors**, **Patients**, and **Appointments**. Department codes are unique. Each doctor belongs to exactly one department; doctor employee numbers are unique within a department. Patients have globally unique patient IDs. An appointment is between exactly one doctor and one patient at a date-time; the same doctor cannot have two appointments at the same date-time. Each appointment may optionally generate one prescription.

Draw the ER diagram with keys and cardinalities, and show how Appointment would be mapped to a relation including an appropriate key.

### Question 2 - Query Languages

[2 pt]

Consider:

*Doctor(DID, Name, Dept)*, *Patient(PID, Name)*, *Appointment(AID, DID, PID, ApptDate, Fee)*, *Prescription*

Write:

- (a) RA to find doctors who have treated at least two different patients.
- (b) TRC to find patients who have never had an appointment costing more than 500.
- (c) SQL to find each department's doctor(s) whose total appointment revenue is maximal in 2026. Keep ties and include doctors with zero 2026 revenue only if the whole department has zero revenue.

### Question 3 - Database Constraints

[1 pt]

Business rule: *A prescription may be inserted only for an existing appointment, and the prescribing doctor must belong to the department that is authorized for that drug.*

Assume *DrugAuthorization(Drug, Dept)*. Give a formal constraint and an SQL-oriented implementation. Distinguish what a foreign key can enforce from what requires a cross-table assertion/trigger.

### Question 4 - Serialization & Concurrency

[2 pt]

Given

$$S = w_1(X) \ r_2(X) \ w_2(Y) \ r_3(Y) \ w_3(X) \ r_1(Y).$$

- (a) Draw the precedence graph and determine conflict serializability.
- (b) If commit order is  $c_1, c_2, c_3$ , determine recoverability and cascadelessness.
- (c) Under basic 2PL, can a deadlock arise if  $T_1$  locks X then requests Y, while  $T_2$  locks Y then requests X? Draw the wait-for edges and give one valid resolution policy.

**Question 5 - Recovery****[1 pt]**

A database uses **REDO-only deferred update**. At crash:

```
<start ckpt (T1,T2)>
<T1, A, 15>
<commit T1>
<T2, B, 22>
<end ckpt>
<T3, C, 31>
<commit T3>
<T2, D, 45>
--- crash ---
```

Explain which transactions are redone, which are ignored, and why no UNDO is required.

**Question 6 - Functional Dependencies & Normalization****[1 pt]**

For  $R(A, B, C, D, E)$ , let

$$F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A, A \rightarrow E\}.$$

- Find all candidate keys.
- Determine the highest normal form of  $R$ .
- For decomposition  $R_1(A, C, D, E)$  and  $R_2(A, B)$ , test lossless join using the common-attribute criterion.

**Question 7 - Concurrency Control Execution****[1 pt]**

Bank balance  $B = 1000$ .

```
T1: r(B); B:=B-100; w(B); commit
T2: r(B); B:=B+50; w(B); abort
```

Assume immediate update and no strictness. Construct one interleaving in which T1 reads a value written by T2 before T2 aborts. State the anomaly and the extra recovery consequence for T1.

**Question 8 - Query Optimization****[1 pt]**

Patient has 500,000 rows, 50 rows/page. There are 10,000 pages. A B+ tree on BirthDate has height 4 and is clustered; a hash index on PID has expected 2 I/Os. Estimate the I/O cost for:

- $PID = 12345$  using hash vs full scan;
- a date range expected to return 5% of rows using the clustered B+ tree vs full scan.

Which index type is unsuitable for the range query, and why?